

END-SEMESTER EXAMINATION, 2023-24
B. Sc. (Hons.) (SEMESTER : 04)
BDA 202 : DATABASE MANAGEMENT SYSTEMS

Time : 03 Hrs.

Max. Marks : 100

- Note:** 1. All questions are compulsory.
 2. Assume missing data suitably, if any.

- CO1 Discuss The Basics Of Databases And Transactions And Data Models. (K1, K2, K3)
 CO2 Discuss About Database Design ,ER-Diagram And Unified Modeling Language. (K1, K3)
 CO3 Explain Relational Algebra And Calculus, Describe Domain Relational Calculus, Calculus Vs Algebra, Computational Capabilities. (K3, K4)
 CO4 Explain And Illustrate Constraints, Views And SQL. (K3,K6)
 CO5 Evaluate Different Types Of Transaction Management. (K4,K5)
 CO6 Explain Concurrency Control, Time Stamping Methods, Optimistic Methods, Database Recovery Management. (K2, K4, K5)

COs Marks BTL

SECTION-A

All Questions are Compulsory:**(10×4=40 Marks)**

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| 1. What are the responsibilities of Database Administrator? | CO1 | 4 | K1 |
| 2. What is data independence? Explain with an example. | CO1 | 4 | K2 |
| 3. What do you mean by cardinality? What are different kinds of cardinalities? | CO2 | 4 | K1 |
| 4. Compare between entity integrity, referential integrity and domain integrity. | CO2 | 4 | K2 |
| 5. Explain the advantages of relational algebra. | CO3 | 4 | K2 |
| 6. Construct Loss less join decomposition with an example. | CO3 | 4 | K3 |
| 7. What is trigger? Explain with an example. | CO4 | 4 | K2 |
| 8. Identify data independence with suitable example. | CO4 | 4 | K3 |
| 9. Explain concurrency. What are the challenges associated with concurrency? | CO5 | 4 | K2 |
| 10. Identify any one deadlock prevention mechanism you have studied. | CO5 | 4 | K3 |

SECTION-B

All Questions are Compulsory:**(3×6=18 Marks)**

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| 11. (a) Categorise the operations performed in relational calculus and relational algebra. | CO3 | 6 | K4 |
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| (b) Analyse the operator used in relational algebra on sets. | | | |
| 12. (a) Analyse the uses of nested sub queries with suitable example. | CO4 | 6 | K4 |

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| (b) Examine the importance of dealing Null values with an example. | | | |
| 13. (a) Assess the requirement of serializability in concurrent transactions with example. | CO5 | 6 | K4 |

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| (b) Explain ACID properties of transaction processing system. | | | |
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SECTION-C

All Questions are Compulsory:

(3×10=30 Marks)

14. (a) Define two tables with one attribute common to both the tables. Now perform join operation on the tables and show the results. State the inference you have drawn from the operation. CO3 10 K4/K5

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- (b) Given the following two relations:

teams(teamid, name) players(playerid, name, teamid, position)

Discover the relational algebra expression for following questions:

1. Find the name and playerid of every player that plays the “center” position.
 2. Find the name of every player that plays on the “Warriors”. How would this expression change if we renamed players’ teamid column to pteamid?
 3. Finds the teamid of all teams that do not have any players.
 4. Write an expression that is equivalent to the following SQL query:
SELECT teamid AS t i d FROM p l a y e r s WHERE pl a y e r s . teamid NOT IN (SELECT teamid FROM teams) AND position=’ shooting guard’;
15. (a) Estimate the following relational database schema consisting of the four relation schemas: passenger (pid, pname, pgender, pcity); agency (aid, aname, acity); flight (fid, fdate, time, src, dest); booking (pid, aid, fid, fdate); Write the SQL queries for : i) Get the details about all flights from Chennai to New Delhi. ii) Find the passenger names for passengers who have bookings on at least one flight. CO4 10 K4/K5

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- (b) Inspect the advantages of joined relations in different ways with examples.
16. (a) Compare Time stamping based Locking mechanism for concurrency control with example. CO5 10 K4/K5

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- (b) Analyse the steps of deadlock detection and deadlock recovery mechanisms in detail.

SECTION-D

Attempt the following Question:

(1×12=12 Marks)

17. (a) Evaluate the properties of transaction. Check whether the schedules are conflict serializable or not and Find out possible serial schedules also. CO6 12 K5/K6
S : R2(A); R3(C); W3(A); W2(A); W2(B); W3(C); R1(A); R1(B); W1(A); W1(B).

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- (b) What is two phase locking mechanisms? Explain the difference between all type of two phase locking scheme.

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